

Recitation: Rational Functions Solutions

Problem 1. Consider the rational function given below:

$$f(x) = \frac{x^2 + 10x + 25}{x^2 + 8x + 15} = \frac{(x+5)^2}{(x+3)(x+5)} = \frac{(x+5)^{\cancel{2}}}{(x+3)\cancel{(x+5)}} = \underbrace{\frac{x+5}{x+3}}_{\text{reduced}}$$

Complete the following parts:

(a) Find the domain.

The domain is where the original denominator is not 0. Solving $(x+3)(x+5) = 0$, we see $x = -5, -3$. Therefore, the domain is $\mathbb{R} \setminus \{-5, -3\}$, i.e. $(-\infty, -5) \cup (-5, -3) \cup (-3, \infty)$.

(b) Find any y -intercepts.

The y -intercept will be the value when $x = 0$. We have $f(0) = \frac{0+0+25}{0+0+15} = \frac{25}{15} = \frac{5}{3}$. Therefore, the y -intercept is $\frac{5}{3}$.

(c) Find any x -intercepts.

The x -intercepts are the solutions to $f(x) = 0$. For a rational function, this will be when the numerator is 0. Setting $(x+5)^2 = 0$, we have $x = -5$. But this value is not in the domain. Therefore, there are no x -intercepts.

(d) Find any vertical asymptotes.

The vertical asymptotes will occur when the denominator is 0 in the ‘reduced’ rational function. So, we solve $x+3 = 0$, which implies $x = -3$. Therefore, the only vertical asymptote is $x = -3$.

(e) Find any horizontal asymptotes.

Examining the original rational function, the numerator has degree 2 and the denominator has degree 2. Therefore, the horizontal asymptote will be the ratio of the leading coefficients, which is $\frac{1}{1} = 1$. Therefore, $y = 1$ is the only horizontal asymptote.

(f) Find any holes.

To find holes, we examine values not in the domain where the denominator is not 0 in the ‘reduced’ rational function. The only such value here is $x = -5$. Therefore, the only hole is at $x = -5$. The y -value will be the y -value at this x -value in the ‘reduced’ rational function. This is $\frac{-5+5}{-5+3} = 0$. Therefore, the hole is $(-5, 0)$ —the ‘missing’ x -intercept!

Problem 2. Consider the rational function given below:

$$g(x) = \frac{2x^3 + x^2 - x}{x^3 + 2x^2 - 3x} = \boxed{\frac{x(2x-1)(x+1)}{x(x+3)(x-1)}} \quad , \quad = \quad , \quad \cancel{x}(2x-1)(x+1) = \underbrace{\frac{(2x-1)(x+1)}{(x+3)(x-1)}}_{\text{reduced}}$$

Complete the following parts:

- (a) Find the domain.

The domain is where the original denominator is not 0. Solving $x(x+3)(x-1) = 0$, we see $x = -3, 0, 1$. Therefore, the domain is $\mathbb{R} \setminus \{-3, 0, 1\}$, i.e. $(-\infty, -3) \cup (-3, 0) \cup (0, 1) \cup (1, \infty)$.

- (b) Find any y -intercepts.

The y -intercept will be the value when $x = 0$. But this value is not in the domain! Therefore, there are no y -intercepts.

- (c) Find any x -intercepts.

The x -intercepts are the solutions to $f(x) = 0$. For a rational function, this will be when the numerator is 0. Setting $x(2x-1)(x+1) = 0$, we have $x = -1, 0, \frac{1}{2}$. But $x = 0$ is not in the domain, and so it cannot be a 0. Therefore, the only x -intercepts are $x = -1, \frac{1}{2}$.

- (d) Find any vertical asymptotes.

The vertical asymptotes will occur when the denominator is 0 in the ‘reduced’ rational function. So, we solve $(x+3)(x-1) = 0$, which implies $x = -3, 1$. Therefore, the vertical asymptotes are $x = -3$ and $x = 1$.

- (e) Find any horizontal asymptotes.

Examining the original rational function, the numerator has degree 3 and the denominator has degree 3. Therefore, the horizontal asymptote will be the ratio of the leading coefficients, which is $\frac{2}{1} = 2$. Therefore, $y = 2$ is the only horizontal asymptote.

- (f) Find any holes.

To find holes, we examine values not in the domain where the denominator is not 0 in the ‘reduced’ rational function. The only such value here is $x = 0$. Therefore, the only hole is at $x = 0$. The y -value will be the y -value at this x -value in the ‘reduced’ rational function. This is $\frac{(0-1)(0+1)}{(0+3)(0-1)} = \frac{-1}{-3} = \frac{1}{3}$. Therefore, the hole is $(0, \frac{1}{3})$ —a ‘missing’ x -intercept.

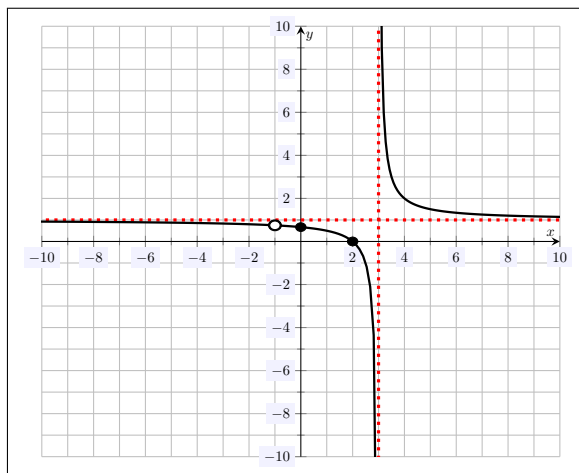
Problem 3. For each of the following, find a rational function with the given properties:

- (a) A function with domain $(-\infty, 5) \cup (5, \infty)$ and a root at $x = 2$.
- (b) A function with a vertical asymptote at $x = 4$ and horizontal asymptote $y = 3$.
- (c) A function with no y -intercept.
- (d) A function with a hole at $(1, -2)$.

Solution.

- (a) *We need to exclude $x = 5$ from the domain, which we could do by forcing the denominator to be 0 at $x = 5$. To have a root at $x = 2$, we could force the numerator to be 0 at $x = 2$. But clearly, $\frac{x-2}{x-5}$ is such a rational function—though there are infinitely many possibilities.*
- (b) *To have a vertical asymptote at $x = 4$, we can force the denominator to be 0 at $x = 4$. To have a horizontal asymptote at $y = 3$, we need the degrees of the numerator and denominator to be the same and the ratio of the leading coefficients to be $3 = \frac{3}{1}$. But clearly, $\frac{3x}{x-4}$ is such a rational function—although there are infinitely many possibilities.*
- (c) *To have no y -intercept, the function cannot be defined at $x = 0$. But clearly, $\frac{1}{x}$ is such a function—although there are infinitely many possibilities.*
- (d) *To have a hole at $(1, -2)$, we need the original rational function to not be defined at $x = 1$ but the ‘reduced’ rational function to be defined at $x = 1$ and have value -2 there. So, the ‘problem’ at $x = 1$ needs to ‘cancel away.’ We can force this by having an $x - 1$ term in the numerator and denominator. But then after reducing, we need the function to have value -2 when we evaluate it at $x = 1$. Clearly, $\frac{-2(x-1)}{x-1}$ is such a rational function.*

Problem 4. Use the plot below to sketch the graph of $f(x) = \frac{x^2 - x - 2}{x^2 - 2x - 3}$.



How do we sketch a rational function? Well, we know how to find ‘all’ the interesting things about it: domain, zeros, asymptotes, holes, etc. So, we can find all these and use this to ‘fill in’ the plot! First, observe...

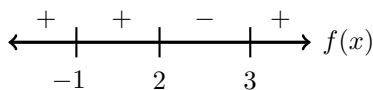
$$\frac{x^2 - x - 2}{x^2 - 2x - 3} = \frac{(x-2)(x+1)}{(x-3)(x+1)}, \quad , \quad \frac{(x-2)\cancel{(x+1)}}{(x-3)\cancel{(x+1)}} = \frac{x-2}{x-3}$$

Domain: The domain is where the original denominator is not 0. Solving $(x-3)(x+1) = 0$, we see $x = -1, 3$. Therefore, the domain is $\mathbb{R} \setminus \{-1, 3\}$.

y-intercept: The y-intercept will be the value when $x = 0$. But we have $f(0) = \frac{0-0-2}{0-0-3} = \frac{-2}{-3} = \frac{2}{3}$. Therefore, the y-intercept is $y = \frac{2}{3}$, so we should be sure to include $(0, \frac{2}{3})$ on our graph!

x-Intercepts: The x-intercepts are the solutions to $f(x) = 0$. For a rational function, this will be when the numerator is 0. Setting $(x-2)(x+1) = 0$, we have $x = -1, 2$. But $x = -1$ is not in the domain, and so it cannot be a 0. Therefore, the only x-intercept is $x = 2$. So, we should include the point $(2, 0)$ on our graph.

Vertical Asymptotes: The vertical asymptotes will occur when the denominator is 0 in the ‘reduced’ rational function. So, we solve $x-3 = 0$, which implies $x = 3$. Therefore, we should sketch a vertical asymptote at $x = 3$. We can even determine how the function approaches this vertical asymptote on the left/right by creating a ‘sign chart’ of the original rational function. We have...



So, the graph should ‘go down to $-\infty$ ’ as it approaches the vertical asymptote on the left, and ‘go up to $+\infty$ ’ as it approaches the vertical asymptote on the right.

Horizontal Asymptote: Examining the original rational function, the numerator has degree 2 and the denominator has degree 2. Therefore, the horizontal asymptote will be the ratio of the leading coefficients, which is $\frac{1}{1} = 1$. Therefore, $y = 1$ is the only horizontal asymptote. We should draw this horizontal asymptote and be sure the function approaches this asymptote as it moves ‘further to the left and right.’

Holes: To find holes, we examine values not in the domain where the denominator is not 0 in the ‘reduced’ rational function. The only such value here is $x = -1$. Therefore, the only hole is at $x = -1$. The y-value will be the y-value at this x-value in the ‘reduced’ rational function. This is $\frac{-1-2}{-1-3} = \frac{-3}{-4} = \frac{3}{4}$. Therefore, the hole is $(-1, \frac{3}{4})$, which we should include in our graph.

Putting all these features together gives us the plot above.